

Morning Herald

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COVER STORY

Bean Machine Retrospective, part 7

How do we write a compiler in a typical general-purpose line-of-business OO programming language such as Python, C#, Java, and so on?

ericlippert · Fabulous Adventures in Coding (Eric Lippert)

Compilers are programs, so we could make the question more general: how do we write programs ?

Divide the problem into many sub-problems

Write functions that each solve one or more sub-problems

Compose a solution by writing functions that call other functions

The details of how those functions are organized varies from language to language of course; functions are stored in other functions, or in classes, or in modules, or whatever. But ultimately, most programs could be viewed as a composition of functions.

Call `foo()` if and only if some predicate is true.

Call `foo()` repeatedly until some predicate is false.

Call `foo()` but branch to this catch block if it fails

... and so on

We don't normally think of control flow as a kind of function composition . What if we did? We can use ideas inspired by combinatory logic and functional programming to extract control flow into "combinators" and then use those to concisely build workflows to solve compiler problems.

A "parse tree" or abstract syntax tree (henceforth AST) is a data structure representing a syntactic analysis of a program. Over the next few episodes of this series we'll explore the question of how a compiler writer might solve a common sub-problem in compiler design: how do we write an `AST → AST` function using an approach inspired by combinatory logic?

Since Bean Machine and its compiler are both written in Python, we'll use the very convenient parse tree types already provided by the Python `ast` module. It's very straightforward. Every node in the tree has a type and zero or more labeled children . A child can be a value such as a string or number, or a node, depending on the label.

For example, if we had a statement `"x = 2 + 3"` then the AST for the right side of the assignment could be constructed like this (assuming that all members of the `ast` module are brought into scope.)

The expression's AST is a binary operator; it has three children, `left` , `right` and `op` . The `left` and `right` are literal numbers; their `child n` is the value of that number. You get the idea I'm sure.

Every Python expression, statement, and so on has an AST node, and there are standard implementations of both parsers and unparsers; you can turn text into ASTs, turn ASTs back into text, and compile and run that program.

Next time on FAIC : I'll describe the patterns/rules/combinators system briefly, and then give some thoughts on what motivated this approach over a more conventional compiler technique such as visitor patterns for rewrites. Then we'll start looking at examples of patterns and predicates.

class="wp-block-paragraph">A “parse tree” or abstract syntax tree (henceforth AST) is a data structure representing a syntactic analysis of a program.

ericlippert

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FRONT PAGE

Travel without social praise

Derek Sivers

I met a couple who were thinking of quitting their jobs and travelling the world for a year. They asked my thoughts.

I suggested they should only do it if they don't bring a camera, and don't tell anyone but their family and few dear friends. No sharing on social media.

Why? Because we often live for others, without even realizing it. We are trying to impress an invisible crowd. We like the social reward of saying, “We're travelling the world!” We imagine how friends and strangers would react to this big news.

We go places we think would be impressive to other people. We take photos that will make our life look wonderful when we share them. We want that praise — that social reward.

Do we really want to do this thing, for its own sake?

Or do we just want the praise?

One way to find out is to see how appealing it would be to do it with no photos, and no sharing. Like the first person to run a marathon without talking about it .

If doing something makes us that kind of person, is it still true if we don't announce it?

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FEATURES

How To Measure The Impact Of Features

Vitaly Friedman · Smashing Magazine

So we design and ship a shiny new feature . How do we know if it's working? How do we measure and track its impact? There is no shortage in UX metrics , but what if we wanted to establish a simple, repeatable , meaningful UX metric — specifically for our features? Well, let's see how to do just that.

I first heard about the TARS framework from Adrian H. Raudschl's wonderful article on “ How To Measure Impact of Features ”. Here, Adrian highlighted how his team tracks and decides which features to focus on — and then maps them against each other in a 2×2 quadrants matrix .

It turned out to be a very useful framework to visualize the impact of UX work through the lens of business metrics.

Let's see how it works.

1. Target Audience (%)

We start by quantifying the target audience by exploring what percentage of a product's users have the specific problem that a feature aims to solve. We can study existing or similar features that try to solve similar problems, and how many users engage with them.

Target audience isn't the same as feature usage though. As Adrian noted, if we know that an existing Export Button feature is used by 5% of all users, it doesn't mean that the target audience is 5%. More users might have the problem that the export feature is trying to solve, but they can't find it.

Question we ask: “What percentage of all our product's users have that specific problem that a new feature aims to solve?”

2. A = Adoption (%)

Next, we measure how well we are “acquiring” our target audience. For that, we track how many users actually engage successfully with that feature over a specific period of time.

We don't focus on CTRs or session duration there, but rather if users meaningfully engage with it. For example, if anything signals that they found it valuable, such as sharing the export URL, the number of exported files, or the usage of filters and settings.

High feature adoption (>60%) suggests that the problem was impactful. Low adoption (50% retention rate (avg.)), we can be quite confident that it has a high strategic importance . A 25–35% retention rate signals medium strategic significance, and retention of 10–20% is then low strategic importance.

Question we ask: “Of all the users who meaningfully adopted a feature, how many came back to use it again?”

4. Satisfaction Score (CES)

Finally, we measure the level of satisfaction that users have with that feature that we've shipped. We don't ask everyone — we ask only “retained” users. It helps us spot hidden troubles that might not be reflected in the retention score.

Once users actually used a feature multiple times, we ask them how easy it was to solve a problem after they used that feature — between “much more difficult” and “much easier than expected”. We know how we want to score.

Using TARS For Feature Strategy

Once we start measuring with TARS, we can calculate an S÷T score — the percentage of Satisfied Users ÷ Target Users. It gives us a sense of how well a feature is performing for our intended target audience. Once we do that for every feature, we can map all features across 4 quadrants in a 2×2 matrix .

Overperforming features are worth paying attention to: they have low retention but high satisfaction. It might simply be features that users don't have to use frequently, but when they do, it's extremely effective.

Liability features have high retention but low satisfaction, so perhaps we need to work on them to improve them. And then we can also identify core features and project features — and have a conversation with designers, PMs, and engineers on what we should work on next.

Conversion Rate Is Not a UX Metric

TARS doesn't cover conversion rate, and for a good reason. As Fabian Lenz noted , conversion is often considered to be the ultimate indicator of success — yet in practice it's always very difficult to present a clear connection between smaller design initiatives and big conversion goals.

The truth is that almost everybody on the team is working towards better conversion. An uptick might be connected to many different initiatives — from sales and marketing to web performance boost to seasonal effects to UX initiatives.

UX can, of course, improve conversion, but it's not really a UX metric. Often, people simply can't choose the product they are using. And often a desired business outcome comes out of necessity and struggle, rather than trust and appreciation.

High Conversion Despite Bad UX

As Fabian writes , high conversion rate can happen despite poor UX, because:

Strong brand power pulls people in,

Aggressive but effective urgency tactics ,

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A year of open source vulnerability trends: CVEs, advisories, and malware

Jonathan Evans · The GitHub Blog

GitHub published 4,101 reviewed advisories in 2025. This is the fewest number of reviewed advisories since 2021. Does this mean open source is shipping more secure code? Let's dig into the data to find out.

Fewer advisories reviewed doesn't mean fewer vulnerabilities were reported. The drop is because GitHub reviewed far fewer older vulnerabilities. When you look only at newly reported vulnerabilities from our sources, GitHub actually reviewed 19% more advisories year over year.

So why the change? Quite frankly, we are running out of unreviewed vulnerabilities that are older than the Advisory Database. At the same time, the number of newly reported vulnerabilities hasn't dropped.

The GitHub Advisory Database provides a comprehensive list of known security vulnerabilities and malware affecting open source packages. It was created in 2019, and has since become a vital resource for open source developers.

[Read more in last year's blog post >](#)

It's also worth clarifying that "unreviewed" in the database can be misleading: most advisories marked unreviewed have already been looked at by a curator and found not to affect any package in a supported ecosystem, so they may never be fully reviewed.

This means that you should be receiving fewer brand-new Dependabot alerts about old vulnerabilities.

Note: If you find an unreviewed advisory that affects a supported package, please let us know so we can get it reviewed!

The distribution of ecosystems in advisories reviewed in 2025 is similar to the overall distribution in the database, with the exception of Go. Go is overrepresented in 2025 advisories by 6%. This is largely due to dedicated campaigns to re-examine potentially missing advisories found through an internal review for packages where we had inconsistent coverage.

Rank Common Weakness Enumeration (CWE) Number of 2025 Advisories* Change in Rank from 2024 Change in Rank from the Overall Database

1	CWE-79	672	+0	+0	2	CWE-22	214	+2	+1	3	CWE-863	169	+9	+8	4	CWE-20	154	+1	+1	5	CWE-200	145	-2	-1	6	CWE-400	144	+4	+0	7	CWE-770	136	+7	+10	8	CWE-502	134	+5	+1	9	CWE-94	119	-3	-1	10	CWE-918	103	+5	+8
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* An advisory may have more than one CWE. For example, an advisory might have both CWE-400 and CWE-770. It would then count for both.

As usual, cross-site scripting (CWE-79) is by far the most common vulnerability type. However, there are significant changes in the following areas. Resource exhaustion

(CWE-400 and CWE-770), unsafe deserialization (CWE-502), and server-side request forgery (CWE-918) were unusually common in 2025. CWE-863 ("Incorrect Authorization") saw a significant jump, but that is largely due to reclassification away from CWE-284 ("Improper Access Control") and CWE-285 ("Improper Authorization"), which are higher level CWEs that the CWE program discourages using.

One of the biggest quality improvements in 2025 was more specific, more consistent CWE tagging. Advisories without any CWE dropped 85% (from 452 in 2024 to 65 in 2025). CWE-20 ("Improper Input Validation") is still common, but in prior years it was often the only CWE listed on an advisory.

In 2025, advisories far more often list CWE-20 plus one or more additional CWEs that describe the concrete failure mode. This added specificity makes the data more actionable for triage, prioritization, and remediation.

To find out how to filter Dependabot alerts by CWE, see our documentation on auto-triage rules.

We provide two scoring systems for prioritization:

Common Vulnerability Severity Score (CVSS): Scores how severe the impact of the vulnerability will be

Exploit Prediction Scoring System (EPSS): Provides a measure of how likely the vulnerability will be attacked in the next 30 days and

Combining the two can help you prioritize which vulnerabilities to address to prevent exploitation.

Jonathan Evans

Together, they can give you a head start on your risk assessment process.

As you can see, when considering impact, most vulnerabilities skew moderate to high of the impact range. Low-impact vulnerabilities are likely more common than the CVSS data suggests but are often not considered worth the time and effort for researchers and maintainers to report. The EPSS scores for moderate to high impact vulnerabilities support this decision.

So should you trust the EPSS or CVSS scores? To judge that, let's look at how they match up to vulnerabilities in CISA's Known Exploited Vulnerabilities Catalog. The exploited vulnerabilities are at least scored moderate, and most are critical or high. While CVSS has more of the exploited vulnerabilities as critical, it also has far more vulnerabilities in the range in general. Combining the two can help you prioritize which vulnerabilities to address to prevent exploitation.

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Bean Machine Retrospective, part 9

ericlippert · Fabulous Adventures in Coding (Eric Lippert)

class="wp-block-paragraph">I wanted to implement concise “pattern matching” in Python, a language which unlike C#, F#, Scala, and so on, does not have any pattern matching built in. Logically a pattern is just a predicate: a function which takes a value and returns true if the value “matches” the pattern, false otherwise. The code for this episode is here .

class="wp-block-paragraph">When I embarked on this I realized two things. First, that it might be nice for debugging purposes to have more information in the return value than just “true” or “false”; in particular, when a complex pattern fails to match, but it should match, then I’ve made a mistake. In order to debug that mistake, I actually made patterns return a new type:

class="wp-block-paragraph">A complex pattern might fail because of its component patterns failing to match, and so that information is included in the failure result. I also made subclasses Success and Fail .

Pattern = Union[PatternBase, int, str, float, type, list, None]

an instance of pattern base

an None, integer, float or string value; the semantics are “the pattern matches if test is equal to the given value”.

a type; the semantics are “the pattern matches if test is of the given type”

a list of patterns; the semantics are “the pattern matches if test is a list of the same length, and each list element matches the corresponding pattern”

class="wp-block-syntaxhighlighter-code "> p = [BinOp, 3, None] # p matches a list of three items where the first is a binary op AST, # the second is the number 3, and the third is the value None

class="wp-block-paragraph">But how would we express “match instances of Foo whose bar attribute is 3”? We can start by solving the more general problem of how do we match against multiple patterns? Here’s our first combinator:

class="wp-block-syntaxhighlighter-code "> def match_every(*patterns: Pattern) -> Pattern: # ... the implementation is tedious to make it efficient, # but you get the idea; we produce a pattern which succeeds if # every given subpattern succeeds, and fails otherwise.

class="wp-block-syntaxhighlighter-code "> class AttributeSubpattern(PatternBase): name: str subpattern: Pattern
def match(self, test: Any) -> MatchResult: submatch = match(self.subpattern, getattr(test, self.name, None)) submatches = {self.name: submatch} if submatch.is_success():
}

return Success(test, submatches) return Fail(test, submatches) # ...

class="wp-block-paragraph">It’s a little clunky to call AttributeSubpattern so I made a synonym function attribute as well. Notice that this class is a combinator; it takes a pattern and produces a new pattern.

class="wp-block-paragraph">And now we can make a third combinator out of the first two: I want a pattern that checks to see if an object is of a particular type, and whether it has an attribute which matches a pattern:

_any = AnyPattern()

def binop(op: Pattern = _any, left: Pattern = _any, right: Pattern = _any) -> Pattern: return type_and_attributes(BinOp, {"op": op, "left": left, "right": right})

is_addition = binop(Add)

Match at least one of these patterns

Match the first item in a list to one pattern and the tail of the list to another

Match at least one item in a list to a given pattern

Match all items in a list to the same pattern

Match the opposite of this pattern

Helper combinators for commonly used AST nodes such as identifiers, operators, and so on.

And so on; you get the idea.

class="wp-block-paragraph">Patterns identify what code needs to be transformed; rules transform them. Next time on FAIC we’ll look at building a set of rules and rule combinators for concisely representing AST transformations.

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Fresh Energy In March (2026 Wallpapers Edition)

Cosima Mielke · Smashing Magazine

Blooming flowers, longer days, milder temperatures — with March just around the corner, the world is slowly but surely awakening from its winter slumber, fueling us with fresh energy. And even if spring is far away in your part of the world, you might sense that 2026 has gained full speed by now, making it the perfect moment to turn those plans and ideas you've been carrying around into action.

To accompany you on all those adventures that March may bring, we have a new collection of desktop wallpapers for you, just as it has been a monthly tradition here at Smashing Magazine for more than 14 years already. Designed by artists and designers from across the globe, each wallpaper comes in a variety of screen resolutions and can be downloaded for free. A huge thank-you to everyone who shared their designs with us — this post wouldn't be possible without your kind support!

If you, too, would like to get featured in one of our upcoming wallpapers editions, please don't hesitate to submit your design. We can't wait to see what you'll come up with! Happy March!

You can click on every image to see a larger preview.

We respect and carefully consider the ideas and motivation behind each and every artist's work. This is why we give all artists the full freedom to explore their creativity and express emotions and experience through their works. This is also why the themes of the wallpapers weren't anyhow influenced by us but rather designed from scratch by the artists themselves.

"With Spring knocking and other seasons fighting to get attention, March greets us with blossoms." — Designed by Ginger It Solutions from Serbia.

preview

with calendar: 320x480, 640x480, 800x480, 800x600, 1024x768, 1024x1024, 1152x864, 1280x720, 1280x800, 1280x960, 1280x1020, 1400x1050, 1440x900, 1600x1200, 1680x1050, 1680x1200, 1920x1080, 1920x1200, 1920x1440, 2560x1440

without calendar: 320x480, 640x480, 800x480, 800x600, 1024x768, 1024x1024, 1152x864, 1280x720, 1280x800, 1280x960, 1280x1020, 1400x1050, 1440x900, 1600x1200, 1680x1050, 1680x1200, 1920x1080, 1920x1200, 1920x1440, 2560x1440

preview

with calendar: 640x480, 800x480, 800x600, 1024x768, 1024x1024, 1152x864, 1280x720, 1280x800, 1280x960, 1280x1024, 1366x768, 1400x1050, 1440x900, 1600x1200, 1680x1050, 1680x1200, 1920x1080, 1920x1200, 1920x1440, 2560x1440, 3840x2160

without calendar: 640x480, 800x480, 800x600, 1024x768, 1024x1024, 1152x864, 1280x720, 1280x800, 1280x960, 1280x1024, 1366x768, 1400x1050, 1440x900, 1600x1200, 1680x1050, 1680x1200, 1920x1080, 1920x1200, 1920x1440, 2560x1440, 3840x2160

I'm Not Okay, But It's Okay

preview

with calendar: 640x480, 800x480, 800x600, 1024x768, 1024x1024, 1152x864, 1280x720, 1280x800, 1280x960, 1280x1024, 1366x768, 1400x1050, 1440x900, 1600x1200, 1680x1050, 1680x1200, 1920x1080, 1920x1200, 1920x1440, 2560x1440, 3840x2160

without calendar: 640x480, 800x480, 800x600, 1024x768, 1024x1024, 1152x864, 1280x720, 1280x800, 1280x960, 1280x1024, 1366x768, 1400x1050, 1440x900, 1600x1200, 1680x1050, 1680x1200, 1920x1080, 1920x1200, 1920x1440, 2560x1440, 3840x2160

Let's Spring

"After some freezing months, it's time to enjoy the sun and flowers. It's party time, colors are coming, so let's spring!" — Designed by Colorsfera from Spain.

preview

without calendar: 320x480, 1024x768, 1024x1024, 1280x800, 1280x960, 1280x1024, 1400x1050, 1440x900, 1600x1200, 1680x1050, 1680x1200, 1920x1080, 1920x1200, 1920x1440, 2560x1440

"This March, our calendar design epitomizes the heralds of spring. Soon enough, you'll be waking up to the singing of swallows, in a room full of sunshine, filled with the empowering smell of daffodil, the first springtime flowers. Spring is the time of rebirth and new beginnings, creativity and inspiration, self-awareness, and inner reflection. Have a budding, thriving spring!" — Designed by PopArt Studio from Serbia.

preview

without calendar: 320x480, 640x480, 800x480, 800x600, 1024x768, 1024x1024, 1152x864, 1280x720, 1280x800, 1280x960, 1280x1024, 1366x768, 1440x900, 1440x1050, 1600x1200, 1680x1050, 1680x1200, 1920x1080, 1920x1200, 1920x1440, 2560x1440

"This month, I want to go to the woods and explore my new world in sunny weather." — Designed by Zi-Cing Hong from Taiwan.

preview

without calendar: 1024x768, 1152x864, 1280x720, 1280x800, 1280x960, 1366x768, 1400x1050, 1440x900, 1600x1200, 1680x1050, 1920x1080, 1920x1200, 1920x1440, 2560x1440

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These Weeks in Remacs III

Wilfred Hughes

Time for another Remacs update: lots of contributions, a wide range of features, and even a logo!

id="contributing">Contributing

Since the last update, we've seen contributions from lots of new people. We've added @brotzeit and @shanavas786, bringing us to seven wonderful people who can approve your PRs.

Speaking of PRs, we've merged an amazing 64 pull requests since the last update!

If you're looking for a good feature for your first contribution, @brotzeit has been regularly adding new suggestions under the 'good first issue' label.

id="features">Features

Many Emacs features have now been ported to Rust, with new Rust APIs for accessing elisp datastructures.

Here's an overview of the features that have landed.

Arithmetic : arithmetic, floating point, random number generation (using a Rust RNG!), and comparisons.

Symbols : symbol properties, interned, obarrays unbinding, keywords and indirect symbols.

Checksums : MD5sum (using a Rust MD5 crate!).

Windows : liveness check, type check, overlays and minibuffer, minibuffer check positions and margins.

Processes : accessing, type check, data structures and names.

Buffers : for the current thread, accessing, file names, size and modification.

Point : bobp, bolp, eolp, markers, point-min, point-max forward-point and goto-char.

Hash tables : copying and accessing.

Characters : multibyte conversions, character tables, category tables

Fonts : type checks.

Miscellaneous : prefix arguments and identity.

We're also periodically pulling GNU Emacs features into Remacs, so all the features available GNU Emacs trunk are included in Remacs.

id="idiomatic-rust-in-remacs">Idiomatic Rust in Remacs

Remacs has gradually developed a set of conventions for elisp data types. For each type Foo, we define a LispObject::as_foo, LispObject::as_foo_or_error and a FooRef when you know your elisp datatype is actually a Foo.

For example, here's how overlay-start was implemented in C:

```
DEFUN ( "overlay-start", Foverlay_start, Soverlay_start, 1,
1, 0, doc : /* Return the position at which OVERLAY starts.
*/ ) ( Lisp_Object overlay ) { CHECK_OVERLAY ( overlay );

return ( Fmarker_position ( OVERLAY_START ( overlay ) ) ); }
```

The C codebase makes heavy use of macros for checking types (CHECK_OVERLAY) and for accessing struct attributes (OVERLAY_START).

Here's the Rust equivalent:

```
/// Return the position at which OVERLAY starts. #[lisp_fn]
fn overlay_start ( overlay : LispObject ) -> LispObject
{ let marker = overlay .as_overlay_or_error () .start ();
marker_position ( marker ) }
```

We use procedural macros to simplify defining an elisp primitive function, and type checking is much more explicit.

(This example is from PR #298.)

Other exciting Rusty features include variadic macros to replace call1, call2 in C with just call! in Rust, and the ability to mock extern C functions so we can write unit tests.

id="hash-maps">Hash Maps

We're not always able to leverage the Rust libraries available. @DavidDeSimone showed some amazing Rust-fu exploring using Rust's FnvHashMap inside Remacs.

Sadly, we weren't able to use the Rust hash map implementation. The C layer assumes that it can mutate hash table keys in place, and unexec does not play nicely with mmap. See the PR for the full details.

id="logo">Logo

Finally, we're discussing a logo for Remacs. We've had some great submissions:

You can join the logo discussion at PR #360.

As always, if you fancy writing some Rust in support of the world's lispier text editor, you can join us on GitHub!

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standard-article(title: [Infinite Scrolling that Works], author: [Robin Ward (eviltrout)], source-name: [Robin Ward (eviltrout)], images: (), paragraphs: ([Shortly after we began working together on Discourse , Jeff wrote a post about infinite scrolling . At first, I was surprised at how many people claimed to hate sites that used it. However, after reading through many comments about it, I realized that most didn't hate the scrolling itself, they hated how it broke their browser!], [id="infinite-scrolling-done-wrong-twitter">Infinite Scrolling done wrong: Twitter], [When I visit Twitter, I am presented with a list of tweets in reverse chronological order. If I scroll down far enough, Twitter will automatically load more tweets so I don't have to stop reading. Initially, their implementation seems great. I can keep scrolling until I'm done reading.],), insert-map: (:), word-count: 110, edited-for-length: false, debug-mode: false,)

standard-article(title: [How and why to make a /now page on your site], author: [Derek Sivers], source-name: [Derek Sivers], images: (), paragraphs: ([I used to wonder what my friend Benny Lewis was doing. He has a website and social media accounts, but neither gave an overview of what he's doing now .], [Then I realized some people might wonder the same about me. So in 2015, I made a /now page on my website, saying what I'd tell a friend I hadn't seen in a year .], [It has a nice side-effect of being a public declaration of priorities. It's a good link to give people when saying no to invitations and distractions.], [Word spread, and soon hundreds of people had a /now page on their personal website . So I made a site to showcase them all: nownownow.com — (a static site generated by PostgreSQL functions .) It currently has over 2300 people worldwide. This week I added browse by location and search.], [Got a personal website? Add a /now page], [The three main ingredients are:], [a page, usually at URL /now, linked from your main menu, usually alongside /about], [an overview of what's going on with you — what you'd tell a friend you hadn't seen in a year], [the date it was last updated], [WordPress instructions:], [In the left menu, under “ Pages ”, click “ Add New Page ”. Then, where it says “Add title”, replace that with just three letters: now . That will ensure the URL is /now, and after it's posted, you can change the title to “What I'm doing now” or whatever.], [Wix instructions:], [On the left, under “ Site Menu ”, click “ + Add Page ”. Call it Now , next to your “About” page. Click the (...) to its right, then to “ SEO basics ”, to edit “ URL slug ” and make sure it's just the three letters: now], [No website yet? Use Bear], [If you don't have a personal website yet, I highly recommend Bear at BearBlog.dev . It's so simple, clean, and free. The owner and creator, Herman in South Africa , runs it himself with great love. And no investors so no enshittification . He plans to keep it alive forever .], [I believe in it so much that I told Herman I would be its godfather. If he ever can't (or doesn't want to) run it anymore, I will help run it, or fund a foundation to keep it alive.], [To create a /now page on Bear, click “ Pages ”, then “ New page ”. Then, after it says “ title: ”, type just three letters: now . That will ensure the URL is /now, and after it's posted, you can change the title to “What I'm doing now” or whatever.], [Got a /now page? I'll add you to nownownow.com], [Once it's live, just email me your URL , and I add it (by hand) to nownownow.com . (This is also a good time to say hello, if you haven't yet. I read and reply to every email.)],), insert-map: (:), word-count: 495, edited-for-length: false, debug-mode: false,)

This week I added browse by location and search.

Derek Sivers

standard-article(title: [Speech to birdsong conversion], author: [Oona Räisänen], source-name: [Oona Raisanen (windytan)], images: (), paragraphs: ([I had a dream one night where a blackbird was talking in human language. When I woke up there was actually a blackbird singing outside the window. Its inflections were curiously speech-like. The dreaming mind only needed to imagine a bunch of additional harmonics to form phonemes and words. One was

standard-article(title: [No new instructions for the computer], author: [Derek Sivers], source-name: [Derek Sivers], images: (), paragraphs: ([You load the program into the computer, and it begins its calculations. It's computing. It's working hard, and it's going to take some time.], [If you interrupt it with new instructions, it has to begin all over again, because the parameters have changed.

left wondering if speech could be transformed into a blackbird song by isolating one of the harmonics...], [One way to do this would be to:], [Find the instantaneous fundamental frequency and amplitude of the speech. For example, filter the harmonics out and use an FM demodulator to find the frequency. Then find the signal envelope amplitude by AM demodulation.], [Generate a new wave with similar amplitude variations but greatly multiplied in frequency.], [A proof-of-concept script using the Perl-SoX-csdr command-line toolchain is available ([source code here](#)). The result sounds surprisingly blackbird-like. Even the little trills are there, probably as a result of FM noise or maybe vocal fry at the end of sentences. I got the best results by speaking slowly and using exaggerated inflection.], [Someone hinted that the type of intonation used in certain automatic announcements is perfect for this kind of conversion. And it seems to be true! Here, a noise gate and reverb has been added to the result to improve it a little:], [And finally, a piece of sound art where this synthetic blackbird song is mixed with a subtle chord and a forest ambience:], [Think of the possibilities: A simultaneous interpreter for talking to birds. A tool for dubbing talking birds in animation or live theatre. Entertainment for cats.], [What other birds could be done with a voice changer like this? What about croaky birds like a duck or a crow?], [(I talked about this blog post a little on NPR: Here's What 'All Things Considered' Sounds Like — In Blackbird Song)],), insert-map: (:), word-count: 320, edited-for-length: false, debug-mode: false,)

standard-article(title: [Tour -isms], author: [Derek Sivers], source-name: [Derek Sivers], images: (), paragraphs: ([Since I'm living in Europe now, I thought it would be good to tour everywhere in Europe, and get to know it better.], [I'm not into seeing the sights. I don't take photos. What I want is to get to know the mindset, the world-view of each place. The philosophy.], [So I wondered if there's a way to tour philosophies directly. What if, instead of touring places, we toured ideas? Can I tour the "-ism"s?], [My collection of Very Short Introduction books includes:], [Post-colonialism], [Post-modernism], [Post-structuralism], [Now that's my kind of tourism!]),), insert-map: (:), word-count: 143, edited-for-length: false, debug-mode: false,)

standard-article(title: [PostgreSQL example of self-contained stored procedures], author: [Derek Sivers], source-name: [Derek Sivers], images: (), paragraphs: ([First, see my previous article about PostgreSQL functions at [sive.rs/pg](#). That article gave tiny examples, but no finished working code.], [This week, I wrote a shopping cart to sell my books directly from my own site.], [So I took a couple extra hours today to put my code into public view, so anyone can play around with it. See [github.com/sivers/store](#), to browse, download, and try it.], [It's a working self-contained shopping cart / store. It's a very concrete example of using stored procedures to keep all the data logic together in one place. You can use it from JavaScript, Python, Ruby, or any language you want, since all the functionality is in

If you keep giving it new information, it will never finish its job.], [People who tell me they are lost and running in circles have one thing in common: They say they keep listening to podcasts, reading books, watching videos, doing courses — taking in more and more information — and still don't know what to do.], [Consider the computer metaphor for yourself. You've taken in so much information, and heard so many instructions. That's enough input. It's time for output. Run the program. Stop interrupting yourself with new information. Let yourself execute one plan of action, and see it through to fruition.],), insert-map: (:), word-count: 146, edited-for-length: false, debug-mode: false,)

standard-article(title: [Fret, Hedge, React], author: [Richard Beck], source-name: [n+1 Magazine], images: (), paragraphs: ([None of the following receives a substantive assessment in When the Clock Broke : NAFTA, Newt Gingrich, Bill Clinton's presidency, the Republican Party's 1994 "Contract with America," Monica Lewinsky, Hillary Clinton's political career, George W. Bush, the 2000 election, September 11, the war on terror, the 2004 expiration of the federal assault weapons ban, the 2008 global financial crisis, Barack Obama, the Tea Party, the legalization of gay marriage, the presence of the Minutemen on the country's southern border, the rise in mass shootings, Occupy Wall Street, Senator Mitch McConnell's lengthy effort to enable minoritarian rule by packing the courts with right-wing judges, Black Lives Matter, #MeToo, or Citizens United.],), insert-map: (:), word-count: 110, edited-for-length: false, debug-mode: false,)

standard-article(title: [Your heroes show which way you're facing], author: [Derek Sivers], source-name: [Derek Sivers], images: (), paragraphs: ([People with many interests often ask my advice on which industry or career path they should follow.], [Years ago, I felt I was just a programmer and entrepreneur. Yes sometimes I write a tiny blog post sharing what I've learned, but that's just something on the side.], [But something never felt quite right about this. I spend most of my time writing, very little time programming, and hadn't started a business in years. Still, I kept saying I was a programmer and entrepreneur, and felt I should really spend more time doing it.], [But everything changed when I asked myself a question:], ["Who are my heroes?"], [I thought, wrote them down, then realized they were all

the database itself. It works.], [If you have any questions, or want to tell me how stupid I am for doing this, email me .],), insert-map: (:), word-count: 139, edited-for-length: false, debug-mode: false,)

standard-article(title: [Podcast published today], author: [Derek Sivers], source-name: [Derek Sivers], images: (), paragraphs: ([Starting today you can follow my podcast at sive.rs/podcast.rss or listen on the web at sive.rs/podcast .], [Each episode is around two minutes long. They are my posts since September 22nd . 33 episodes so far.], [I generated the RSS XML feed myself using this Ruby script . The MP3s are just hosted on my own server. I skipped all the podcast hosting services, because I'll never have ads so I don't care about analytics, tracking, and all of that.], [It doesn't cost me anything, so I won't be trying to make money from it. I'm doing it just because people keep asking me to. ☺], [This is all an experiment. Please let me know if you have any suggestions.],), insert-map: (:), word-count: 120, edited-for-length: false, debug-mode: false,)

standard-article(title: [Mapping microwave relay links from video], author: [Oona Räisänen], source-name: [Oona

authors ! Basically, look at my list of favorite books , and there are my heroes.], [The people I look up to the most... The people I'd most like to meet... The people I'd most like to emulate are not entrepreneurs, and not programmers — just writers.], [So, that day, I realized I actually want to be a writer.], [I re-arranged my hierarchy of interests. Yes I enjoy programming, and yes I'll probably start another business. But really my main love and top priority is writing.], [How about you? Who are your heroes? Does that help you see which way you're actually facing?],), insert-map: (:), word-count: 220, edited-for-length: false, debug-mode: false,)

standard-article(title: [Why did I move to New Zealand?], author: [Derek Sivers], source-name: [Derek Sivers], images: (), paragraphs: ([I get this question a lot, so here's my honest answer.], [I was living in Singapore, feeling culturally adventurous, wanting to live everywhere, meet everyone, and get to know the whole world. But then I had a baby.], [At first I thought we'd raise him in Singapore, fluent in all the cultures of Asia. But after a few months, I realized how important it was to me that he have a real connection with nature. Feet in the river, hands in the mud, climbing trees, running in fields, at home in the forest. Great foundation for the soul. Maybe even a competitive advantage in a world where everyone else is dependent on devices.], [Also, Singapore was a super-social place for me, where I knew too many people, and I wanted to give my boy my full attention without distraction — to live where I didn't know anyone. So the plan was to raise him in nature for the first six years, then move somewhere more culturally expansive for the next six.], [New Zealand is a nature paradise, but I also loved the safety of its location in times of global trouble. They make you a citizen if you live here for six years. So, perfect match for our situation. Raise a baby in nature, then leave with an amazing passport, and the right to return forever.], [It took nine months of paperwork to become a legal resident, but the government is friendly and helpful. It's really sweet to live in a country where the government agencies are really small, sensible, and personable. I like the culture here better than I expected.], [His upbringing went as planned. Full nature boy. Not into screens or phones at all. He's always outside, spends his days in forests and the rocky coast, and loves building physical things like tools and shelters for his adventures.], [After getting New Zealand citizenship, we moved to Europe, but then Covid hit. So in a way it was just as predicted: New Zealand was a great location in times of global trouble - one of the only Covid-free places on earth - and only citizens were allowed in, so our passports felt like the golden ticket, and we moved back. While his cousins across the world were locked down in isolation, staring at screens, he was playing with friends.], [His mom loves her job in the government here, so no more moving. He's in high school now, and he's still not into screens. He's mentally healthy and wise, I think partially due to being raised outside in nature.], [I've made a lot of mistakes in my life, but moving to New Zealand was a good choice.],), insert-map: (:), word-count: 447, edited-for-length: false, debug-mode: false,)

standard-article(title: [The Siren Song of Little Languages], author: [Wilfred Hughes], source-name: [Wilfred Hughes],

Raisanen (windytan)], images: (), paragraphs: ([Radio networks are often at least partially based on microwave relay links . They're those little mushroom-like appendices growing out of cell towers and building-mounted base stations. Technically, they're carefully directed dish antennas linking such towers together over a line-of-sight connection. I'm collecting a little map of nearby link stations, trying to find out how they're interconnected and which network they belong to.], [Circling around], [We can find a rough direction for any link antenna by approximating a tangent for the dish shroud surface from position-stamped video footage taken while circling the tower. Optimally we would have a drone make a full circle around the tower at a constant distance and elevation to map all antennas at once; but if our DJI Phantom has run out of battery, a GPS positioned still camera at ground level will also do.], [The rest can be done manually, or using Hough transform and centroid calculation from OpenCV. In these pictures, the ratio of the diameters of the concentric circles is a sinusoid function of the angle between the antenna direction and the camera direction. At its maximum, we're looking straight at the beam. (The ratio won't max out at unity in this case, because we're looking at the antenna slightly from below.) We can select the frame with the maximum ratio from high-speed footage, or we can interpolate a smooth sinusoid to get an even better value.], [This particular antenna is pointing west-northwest with an azimuth of 290°.], [What about distance?], [Because of the line-of-sight requirement, we also know the maximum possible distance to the linked tower, using the formula $7140 \times \sqrt{(4 / 3 \times h)}$ where h is the height of the antenna from ground. If the beam happens to hit a previously mapped tower closer than this distance, we can assume they're connected!], [This antenna is communicating to a tower not further away than 48 km. Judging from the building it's standing on, it belongs to a government trunked radio network.],), insert-map: (:), word-count: 338, edited-for-length: false, debug-mode: false,)

images: (), paragraphs: ([Some programming languages languish due to obscurity. They lack breathless blog posts exclaiming how much nicer they are to use.], [Other languages are too ambitious. They aspire to support so many features that the original implementers struggle to get a first version working. For example, the type system in Fortress required constraint solving which took exponential time .], [Sometimes a usable language struggles simply because it's too much fun to write your own . Developers end up building their own implementation rather than actually using the language.], [The most obvious implementation-focused language is BF . Despite having many implementations , BF programmers have to encourage the implementers to actually try using the language !], [Scheme is also susceptible to this. Wikipedia lists 31 different Scheme implementations , not to mention the many toy implementations. Writing a Scheme is a great introduction to interpreters, especially once you get beyond the minimal lisp featureset . I've certainly written more implementation code than Scheme code.], [The problem seems to be languages with a small, well written specification. Shen is a multiparadigm lisp defined in terms of an elegant base language with only 46 system functions. This has resulted in a remarkable 15 third-party implementations , but only a small number of libraries implemented in the language .], [This phenomenon is not limited to lisps. Forth is also a language that developers often prefer to implement rather than use. Jones Forth is both a Forth tutorial and a discussion of how to build a Forth compiler. There are even stories of people spending years working on implementations without learning much of the language .], [Designing a language with a straightforward implementation is not a bad thing. It's just a pitfall that language designers need to be aware of. Some crypto systems have this problem too .], [It seems that we need languages to be big enough that new users write hello world in the language, not write a tool for others to write hello world. In the Lisp family, Clojure and Racket seem to have reached a size threshold where newcomers are happy downloading the canonical implementation.], [This doesn't mean multiple implementations are bad. It's a great sign of language health. We just don't need these until there's a community of users and there is a need for implementations with different qualities.], [A language can be too small and elegant.],), insert-map: (:), word-count: 396, edited-for-length: false, debug-mode: false,)

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ANALYSIS

Elias Rodrigues, n+1 Magazine: Trump was — and is — intent on creating a new future, and to gain a better sense of that vision, I needed to understand what future he was working to prevent. I asked Andil to meet that weekend, and he agreed. I would play amateur journalist and interview Andil again, this time about how he fell into the government's crosshairs.

Maria Marchinkoski, n+1 Magazine: We have no tradition, she said. What could we say to the dead except that we're sorry for living? No, we confuse people too much already. How much more could they take? They hear "transsexual" and want us to prove it. They want weeping in front of mirrors. They want heartfelt confessions with parents on the couch. They will never take you seriously or consider you normal. They want hand-me-down emotions.

Rubin DasguptaChristian SteinmetzgerAncy T. WilsonSatyaki ChatterjeeGunnar W. ReginssonSnorri Th. SigurdssonKatja PetzoldaDepartment of Medical Biochemistry and Microbiology, Science for Life Laboratory, Center of Excellence for the Chemical Mechanisms of Life, Uppsala University, Uppsala 751 23, SwedenbDepartment of Chemistry, Science Institute, University of Iceland, Reykjavik 107, Iceland, PNAS News: Proceedings of the National Academy of Sciences, Volume 123, Issue 12, March 2026. Significance—Understanding RNA structure and dynamics inside cells remains a key challenge in molecular biology. Here, we establish a site-directed spin labeling approach that enables targeted NMR signal enhancement of RNA in complex environments, using ...

Sumitabha BrahmachariAntonio B. OliveiraMatheus F. MelloVinicius G. ContessotoJosé N. OnuchicaCenter for Theoretical Biological Physics, Rice University, Houston, TX 77005bPhD Program in Systems, Synthetic, and Physical Biology, Rice University, Houston, TX 77005cDepartment of Physics and Astronomy, Rice University, Houston, TX 77005dDepartment of Chemistry, Rice University, Houston, TX 77005eDepartment of Biosciences, Rice University, Houston, TX 77005, PNAS News: Proceedings of the National Academy of Sciences, Volume 123, Issue 12, March 2026. SignificanceSegregation of duplicated DNA is essential for cell proliferation. By combining experimental Hi-C contact maps with a energy landscape approach, we uncover how Structural Maintenance of Chromosomes (SMC) complexes reorganize the chromosome ...

Liv Veazey, n+1 Magazine: All the Immigration and Customs Enforcement agents I saw in the atrium were white, of medium height and build, and many wore plain baseball caps. They were all men and they stuck out because they displayed an exceeding level of stillness and homogeneity in a room with a flow of people from all over the world, people who were always in motion and who mostly looked different from one another.

Pallavi SinghMydhily VasudevanSwagata DeyKavinila SelvarasuSupraja BalakrishnanDikshalee BassiAnindya Ghosh-RoyaDepartment of Cellular and Molecular Neuroscience, National Brain Research Centre, Manesar, Haryana 122052, India, PNAS News: Proceedings of the National Academy of Sciences, Volume 123, Issue 12, March 2026. SignificancePrevious studies have established that physical injury to dendrites leads to the initiation of dendrite regeneration by a mechanism independent of pathways involved in sensing axonal injury. Therefore, the mechanism of dendrite regeneration is ...

Andreas MühlbauerYuanbei F. FanDaniel J. SamborMark Z. JacobsonaDepartment of Civil and Environmental Engineering, Stanford University, Stanford, CA 94305-4020, PNAS News: Proceedings of the National Academy of Sciences, Volume 123, Issue 12, March 2026. SignificanceThis study reveals the immense, previously unquantified potential of repowering the United States (US) onshore wind turbine fleet. By replacing aging turbines with modern technology at existing sites, the United States could more than double ...

Melinda Cooper, Malcolm Harris, n+1 Magazine: These conflicts tend to recur every ten years or so, with a different cast of characters, but always involving the idea that taxpayer money is being spent on a public institution that undermines the private authority of parents —abortion clinics, child care centers, public libraries, public schools.

Donald P. McDonnellaDepartment of Pharmacology and Cancer Biology, Duke University School of Medicine, Durham, NC 27710, PNAS News: Proceedings of the National Academy of Sciences, Volume 123, Issue 12, March 2026. Bert W. O'Malley, the "father/grandfather" of the field of molecular endocrinology passed away on November 11, 2025, at the age of 88. In a distinguished career that spanned over 60 years, he probed fundamental aspects of steroid hormone action, work that ...

Eugene MalthouseCharlie PilgrimDaniel SgroiMichela AccerenziaAntonio AlfonsoRana Umair AshrafMax BaardSanchayan BanerjeeAlexis BelianinSwagata BhattacharjeeMihir BhattacharyaPablo Brañas-GarzaJuan-Camilo CárdenasMiguel CarriquirySyn-gjoo ChoiGwen-Jiro ClochardEduardo Ezekiel DenzonBartłomiej Dessoulavy-SliwinskiGiorgio DiniLu DongAntal ErtlFilippos ExadaktylosEmel Filiz-OzbaySarah Lynn FleckeFabio GaleottiTeresa Garcia-MuñozNobuyuki HanakiGuillaume HollardDaniel Horn-Lingbo HuangDoruk İrişHubert Janos KissJuliane KochJaromír KovářikOsbert Kwabena Boadi KwartengAndreas LangeMartin LeitesThomas Ho-Fung LeungWooyoung LimMeike MorrenLaila NockurCharles Yaw OkyereMayada OudahAli I. OzkesLionel Page-Junghyun ParkStefan PfattheicherAntonios ProestakisCarlos RamosMapi Ramos-SosaMuhammad Saeed AshrafMuhammad Ryan SanjayaRene SchwaigerOmar SeneFei SongSarah SpycherRostislav StaněkNorman TanchingcoAlessandro TavoniVera te VeldeMaria José Vázquez-De FranciscoMartine VisserJoseph Tao-Yi WangWilly WangWei-Chien WengKatharina WernerAmanda WijayantiRalph WinklerJohn WoodersLi YingWei ZhenThomas HillsaCentre for Decision Research and Experimental Economics, School of Economics, University of Nottingham, Nottingham NR7 2QX, United KingdombSchool of Mathematics, University of Leeds, Leeds LS2 9JT, United KingdomcDepartment of Economics, University of Warwick, Coventry CV4 7AL, United KingdomdFundación ETEA—Development Institute of Universidad Loyola Andalucía, Córdoba 14004, SpaineDepartment of Applied Economics I, University of Seville, Sevilla 41018, SpainfFaculty of Economic Sciences, University of Warsaw, Warsaw 00-241, PolandgEnvironmental Policy Research Unit, University of Cape Town, Cape Town 7700, South AfricahSchool for Government, The Policy Institute, King's College London, London WC2B 4LL, United KingdomiNational Research University Higher School of Economics, Moscow 101000, RussiajAlmaty Management University, Almaty 050060, KazakhstankJindal School of Government and Public Policy, O.P. Jindal Global University, Haryana 131001, IndiaDepartment of Economics, Ashoka University, Haryana 131029, IndiaLoyolaBehLAB, Department of Finance, Universidad Loyola Andalucía, Córdoba 14004, SpainmFacultad de Economía, Universidad de los Andes, Cundinamarca, Bogotá 111711, ColombiaonDepartment of Economics, University of Massachusetts Amherst, Amherst, MA 01003pInstituto de Economía, Universidad de la República de Uruguay, Montevideo 11200, UruguayqDepartment of Economics, Seoul National University, Seoul 08826, Republic of KorearInstitute of Social and Economics Research, University of Osaka, Osaka 567-0047, JapanJoint Initiative for Latin American Experimental Economics, Buenos Aires C1054 AAF, ArgentinaJohn Gokongwei School of Management, Ateneo de Manila University, Metro Manila 1108, PhilippinessDepartment of Economics, University of Bologna, Bologna 40126, ItalytSouthern University of Science and Technology, Guangdong Province 518055, ChinauELITE Centre for Economic and Regional

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